



Subject - Digital Image Processing

8. Vehicle Orientation Estimation Using Conventional Image Processing

Weekly Report - 1

Date of Submission: 12/09/2026

Repo Link: https://github.com/encrypted-official/DIP_2026_Project8_Group1

Group - 1

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2026 (Monsoon Semester)

1. Work done this week

All of us this week concentrated on understanding the problem statement for our project titled **“Precision based Vehicle Orientation Estimation Using Conventional Image Processing.”** We researched papers related to vehicle orientation, object recognition, and angle estimation.

Conventional methods we noted were PCA, HOG, edge detection, contours, Hough Transform, and Radon Transform. We began checking the research released this July on the DRASHTI-HaOBB database.

2. Work to be done next week

For the next week, we will start working with the DRASHTI-HaOBB dataset (provided in the reference projects) and its vehicle crops. We plan to:

- Convert the given heading angles into the required orientation format.
- Start preprocessing the vehicle images.
- Implement a basic PCA-based orientation estimation method.
- Explore edge and contour-based approaches.
- Test the initial methods on a small set of vehicle images.

3. Literature Review

From the papers we checked, we found that conventional methods could be used to estimate vehicle orientation without using any deep learning.

- Zhou et al. (2017) used Canny edge detection and the Radon Transform to estimate vehicle orientation from aerial images. This is closely related to our project since our dataset also contains aerial vehicle images.
- Bhavsar et al. (2026) demonstrated the use of UAV video analysis for studying vehicles and traffic behaviour in our Indian traffic environment. We noted and tried to understand the usefulness of UAV-based vehicle analysis.
- Rybski et al. (2010) used HOG features for classifying the various orientations of vehicles. This gave us yet another traditional method for feature extraction from vehicle images.
- Ajmal & Hussain (2010) used Morphological Image processing for segment vehicles and used highway images by differencing the references of empty road against current frame, then applied road mask, edge detection and multilevel threshold.

- Hutter & Brewer (2009) match the 2d ellipses to 3d circles with application of vehicle pose identification. They detect car wheels with ellipses and then get each ellipse's axes from blob's covariance matrix and then map these to 3d circles or wheels to know the orientation.

4. Challenges Faced

- Understanding the difference between the vehicle orientation and direction its heading.
- Dealing with different vehicle shapes and sizes.
- Considering the effect of blur, shadows, noise, and/or low-resolution vehicle crops.
- Deciding which conventional method will give the most reliable orientation estimates.